CASCADOS

WATER SCARCITY IN MEXICO CITY 2054

future scenario

IN 2024

- Inefficient public infrastructure
- Government corruption
- Polluted water

- Land subsidence
- Leaky pipes
- Rising energy costs related to aquifer depletion

- Urban reforestation campaigns
- Increased R&D funding for green tech

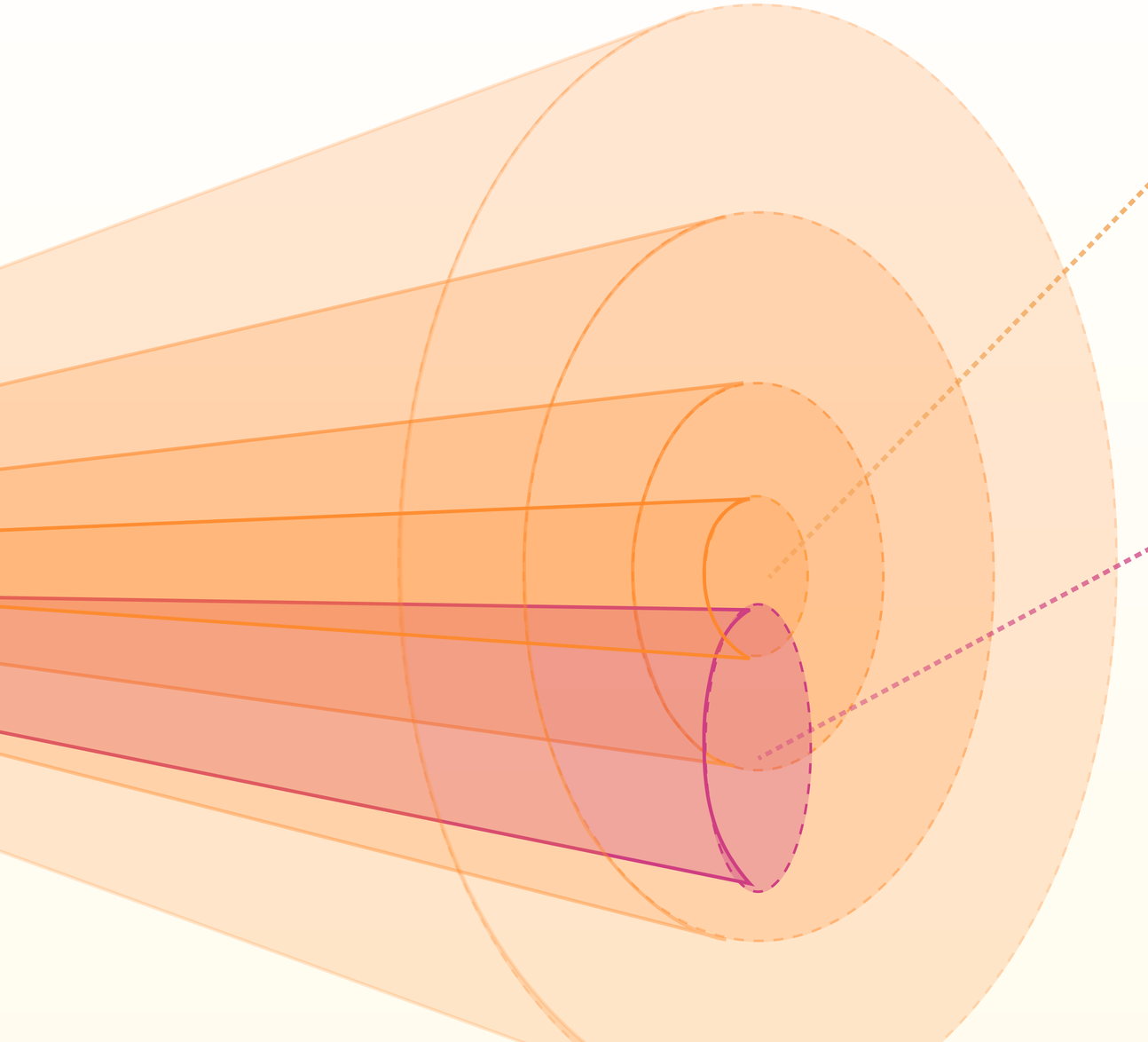
IN 2054

Decentralised water systems, less reliance on public infrastructure

Decrease in groundwater harvesting, prioritising alternative water sources

Integration of nature-based solutions in urban water management

futures cone



PROJECTED FUTURE

Continued reliance on aquifers

Insufficient investment in public infrastructure

PREFERRED & PROBABLE FUTURE

Decentralised water management

Reduced dependence of households on public infrastructure and government plans

Potable water for every household

user expectations 2054

Current source of water:
Limited central water + Water truck and bottles

Affordable technology supporting **consistent**, potable water supply.
More water in cleaner **quality**.

System reliability, efficiency
Water quality

Healthier, cleaner **living conditions**.
Basic needs secured and concerns for **poverty** reduced.

Affordability
Local adaptation

Less reliance on central government, the system operation is **transparent and reliable**.

Decentralised system

How might we purify water locally to decrease the user's reliance on a central water source?

DURABILITY

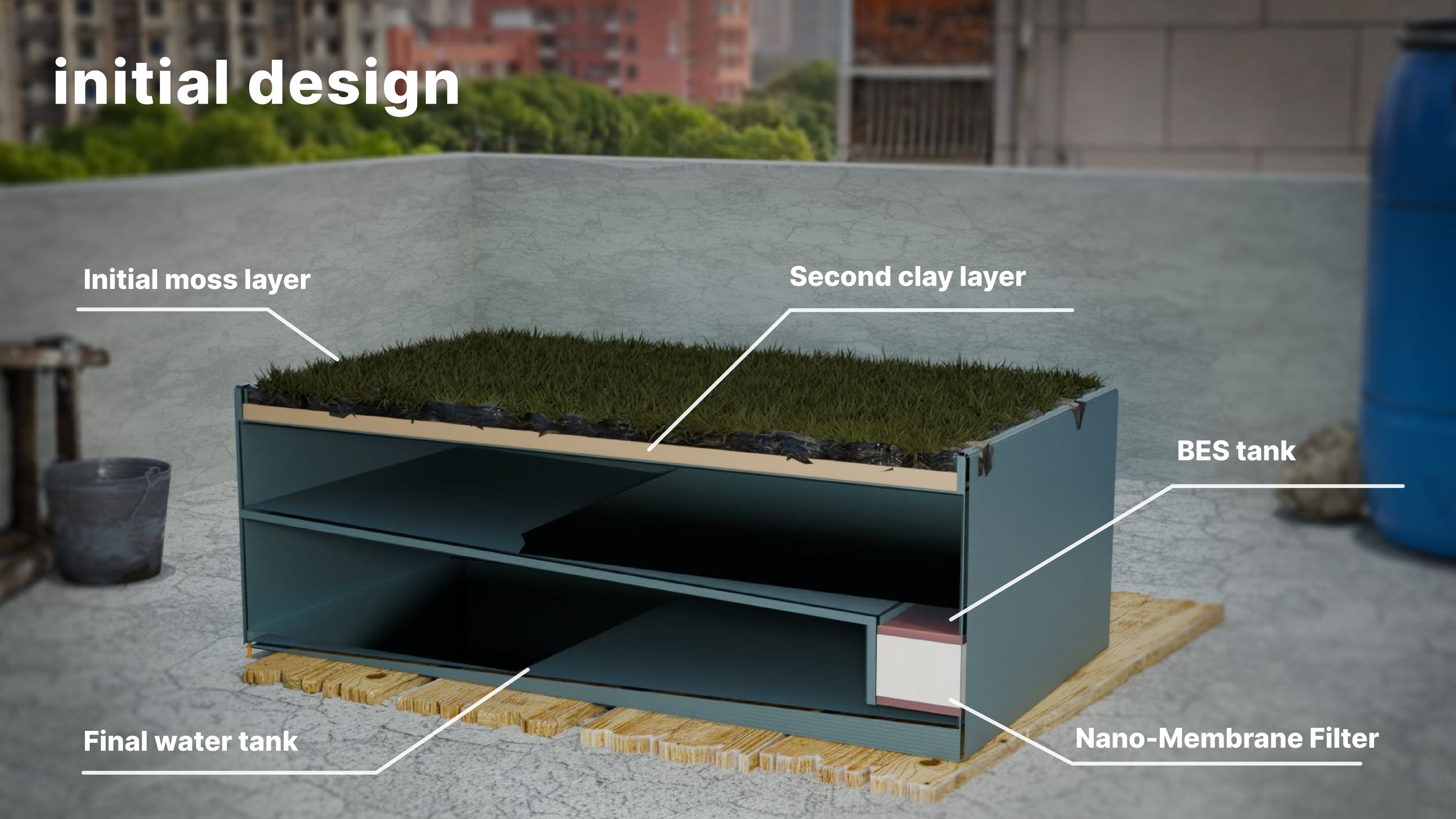
AFFORDABILITY

WATER QUALITY

DECENTRALISATION

EASE OF USE

initial design



TERM 1

Feedbacks

Re-evaluations

WATER SOURCE

- Rainwater
- Greywater
- Blackwater

USER

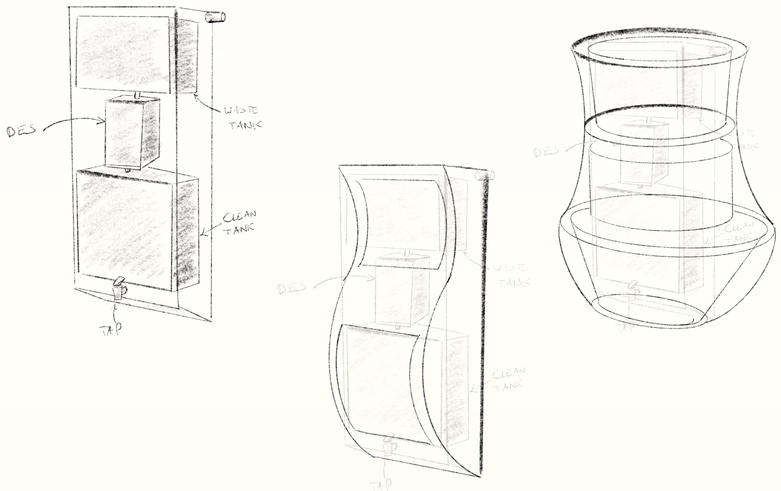
- User needs
- Maintenance

SCALE

- Individual
- Communal

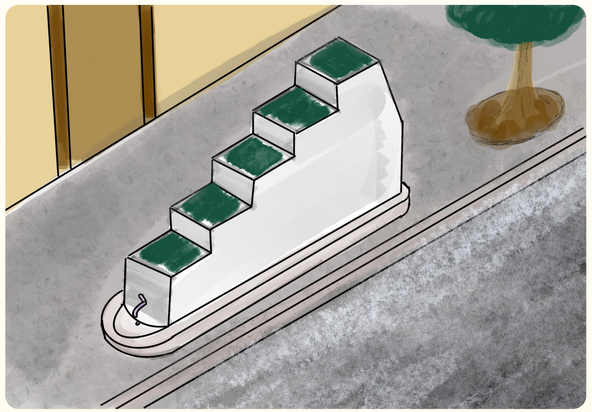
variation of ideas

IN-HOME RECYCLING

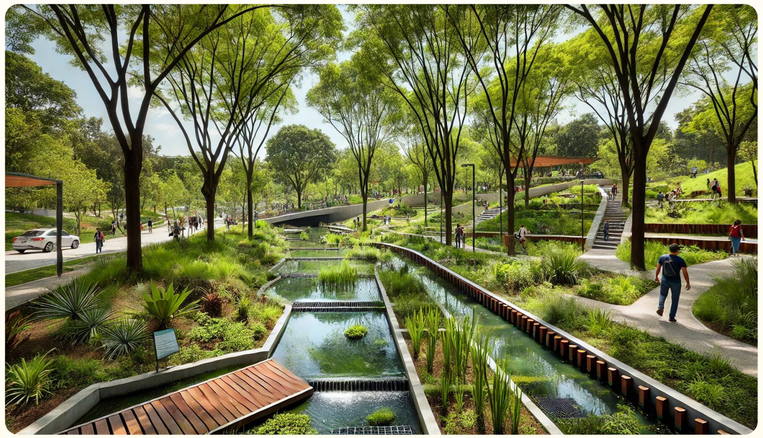


RE-USING BEETLES

COMMUNAL RECYCLING



GREEN PARKS



water source

RAINWATER

Surface area
178,000 m^2

÷

Population
450,000

=

Area available per person **0.4m²**

Annual water usage
150L * 365 days

÷

Annual rainfall
725 L/m^2

=

Area Necessary per person **75m²**

BLACKWATER VS GREYWATER

Black water: sewage from toilets, heavily contaminated

Requires considerable construction for tapping into waste water pipes

Grey water: wastewater from sinks and laundry, reusable with treatment

Can be utilized with minimal integration of in-house pipelines

user interaction

USER NEEDS BEYOND WATER

Health concerns: water quality

Economic constraints

Water quality: Arsenic dissolution

Purification monitoring

Integration with home appliances

REGIONAL PREFERENCES

Irregular supply of tap water →

Operating independently of continuous water flow, with storage capacity

Transparent purification process by visible filtration stages, lab testing results

Decentralized approach, reducing the dependence on failing infrastructure

↓

Successful Local Adoption and Trust

scale

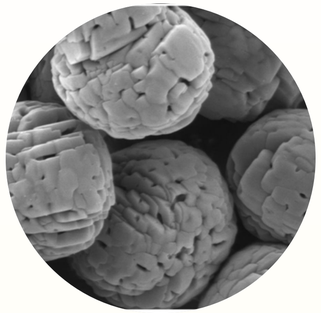
DECENTRALISATION

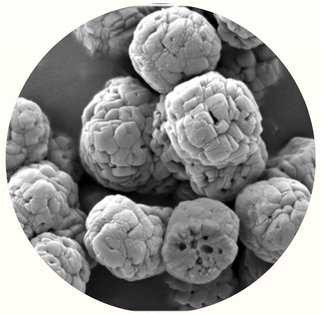
- Mexico city suffers from a poor state of pipeline maintenance, corruption, and unreliable infrastructures.
- These unstable conditions highlight the advantage of an isolated, decentralised system for a safe water source

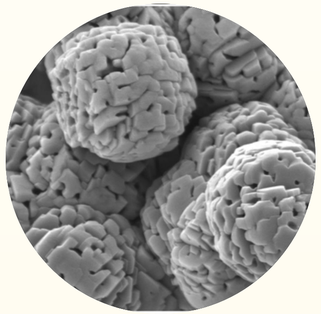
Different scales for implementation:

NEIGHBOURHOOD -SCALE VS INDIVIDUAL VS FULLY OFF-GRID

layer 1. ceramic-zeolite filter







WHAT IS CERAMIC-ZEOLITE FILTRATION?

- Porous Composite Material
- Dual-Action Filtration

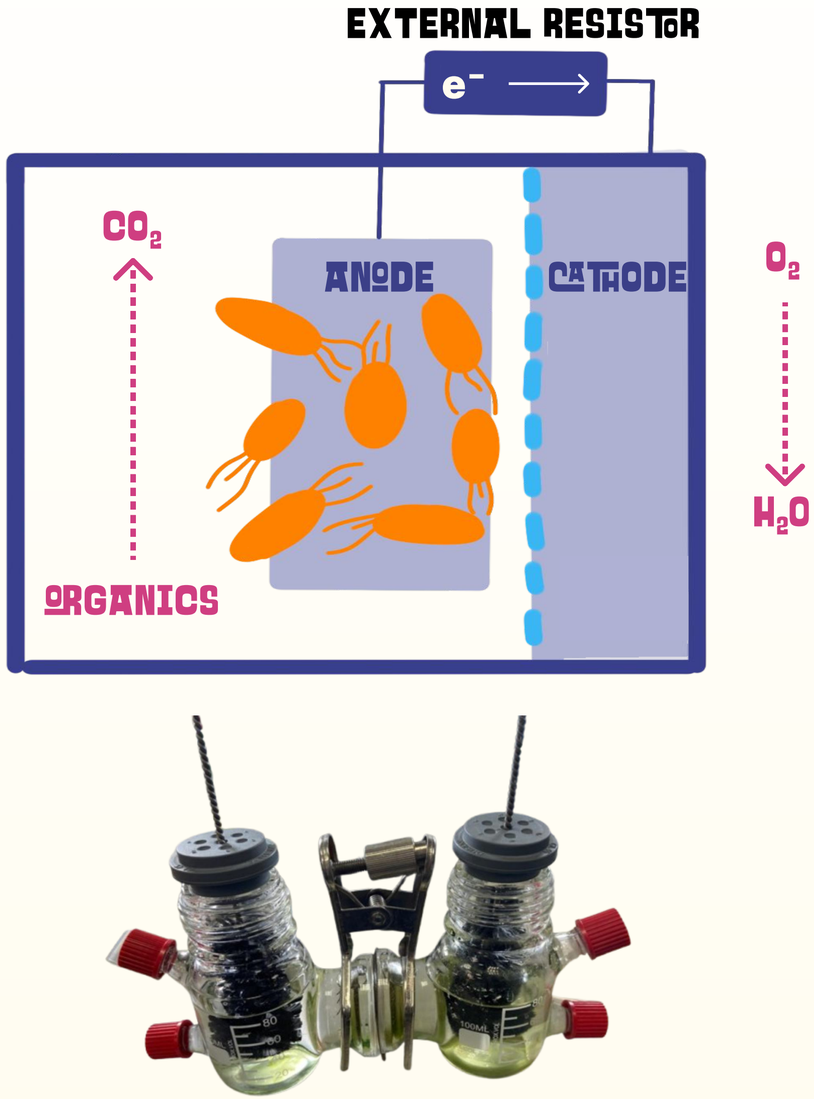
WHY REPLACE CLAY & MOSS LAYERS?

- Short Lifespan
- Limited Filtration Efficiency

HOW DOES CERAMIC-ZEOLITE FILTRATION IMPROVE OVER TRADITIONAL ZEOLITE?

- Precise Pore Size Control
- Improved Filtration Uniformity
- Better Chemical & Thermal Stability

layer 2. microbial fuel cells

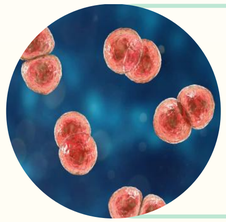


What is Microbial Fuel Cells?

MFC (Microbial Fuel Cell) is a type of BES that generates electricity by breaking down organic matter using bacteria.

Why Microbial Fuel Cells (MFCs)?

- Superior Organic Pollutant Treatment Efficiency
- Highest Power Output Among BES Systems



Why a Single-Chamber MFC?

- Lower Cost with Scalability & Flexibility
- Easier maintenance: anti-fouling spray

How Long Can MFC Microbes Survive?

- Over 10 years in a stable and proper environment
- Several weeks in a humid but water-limited environment

MFC future forecast

IN 2054

Membrane Enhancements

- Membranes for **improved ion selectivity**
- **Self-repairing** and adaptive biofilms

Bioengineering Microorganisms

- **Genetically modified bacteria** to enhance electron transfer efficiently

Electrode Improvements

- **Carbon fibre electrodes:** 50 years lifespan
- **Corrosion-resistant titanium cathodes**

6.25L processed by MFC per hour

140L processed by MFC per day

45-67W power generated

Dr. Tabish Noori

PostDoc in bioelectrochemical process design

> 10x efficiency is plausible in 30 years future.
> [6.25L] suitable amount of water to yield the best result.
> Considering the extensive research on energy maximisation, the power output would be reachable in 2054.

final design

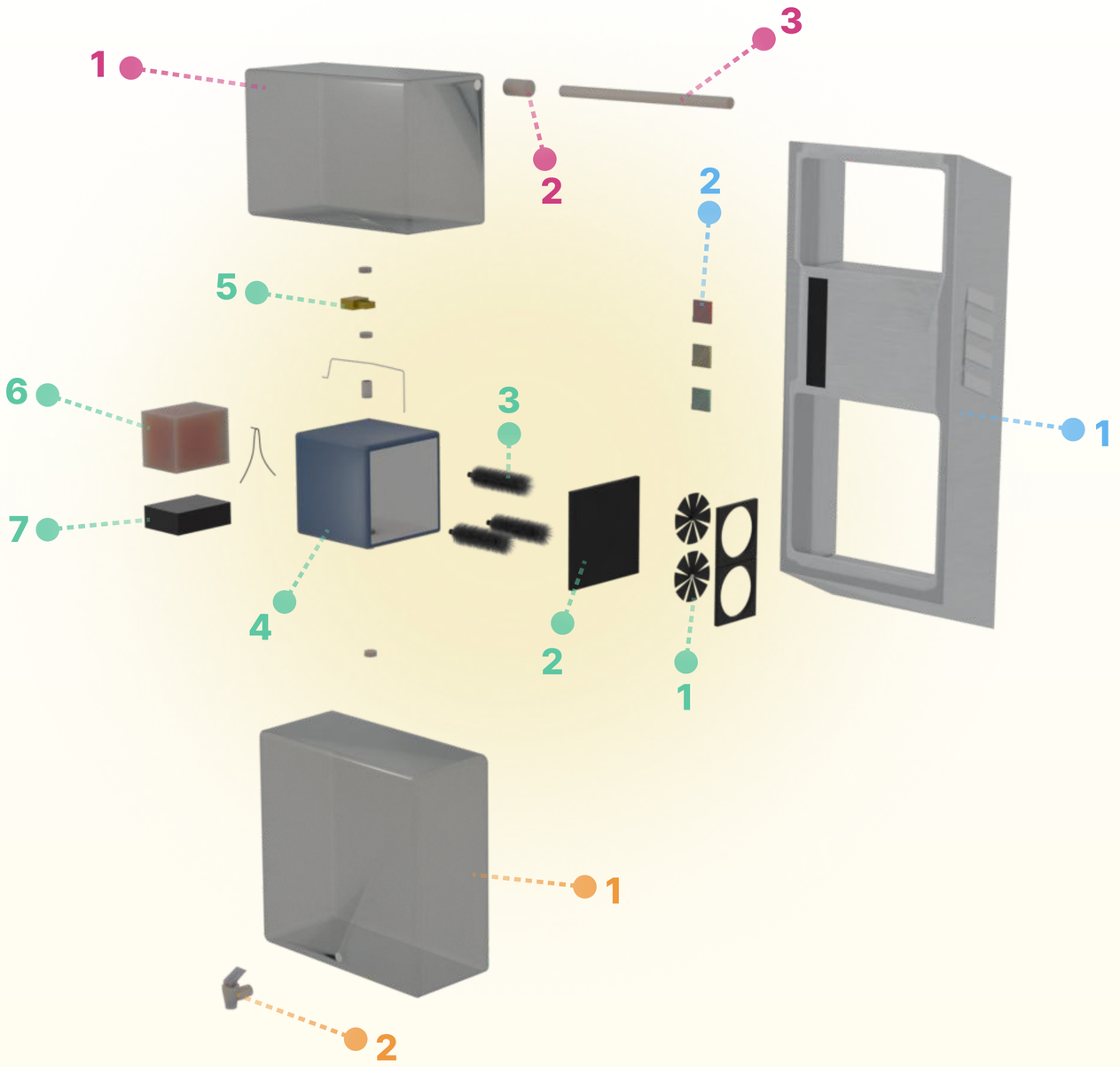
exploded view

TOP TANK

1. Tank
2. Ceramic-Zeolite Filter
3. In Pipe

MFC

1. Fans
2. Cathode & Membrane
3. Carbon Brush Anodes
4. Tank
5. Valve
6. Management Unit
7. Battery



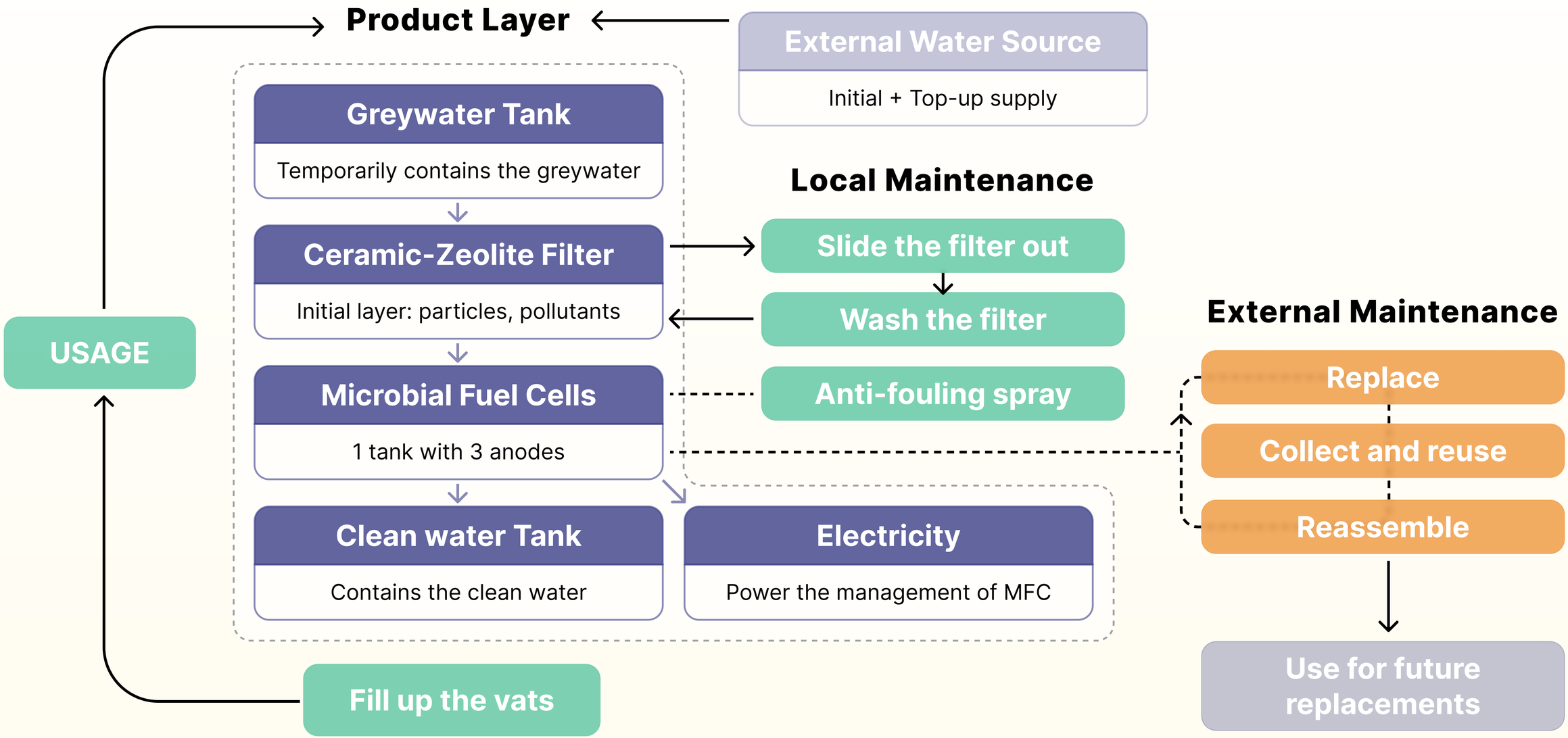
CASING

1. Steel Cover
2. LEDs

BOTTOM TANK

1. Tank
2. Tap

product service system



validation: financial feasibility

$ 4.5 BILLION

National Water Commission (Conagua) budget
Fifteen priority water projects: construction of dams, aqueducts, and irrigation districts

$ 1.44 BILLION

Agua Bienestar initial investment, initiated 2025
Distribution of around 200,000 water bottles per week

584,000 PEOPLE

Number of people non-profit organisation **Isla Urbana** helped with rainwater harvesting

*100,000L/day → **$32760/year** for 10 year life*

*Our system generating **140L/day***

***1.8 Million residents** in Iztapalapa*

$ 960 MILLION

validation: design impact

DESIGN OBJECTIVES

Purification

Visual engagement

Economics

Ease of use

Ease of maintenance

LIMITATIONS

Considerable reliance on mains

Unadaptable for irregular usage

Minimal construction required

Limited DIY maintenance

SOCIAL IMPACT

Health Impact

- Improve water quality and reduce health risks.
- Water access allows for showering and more frequent laundry, **improving hygiene** and quality of life.
- Relieves long lasting **mental stress** on water acquisition

Job Creation

- Generate employment opportunities through **production, distribution**, and **maintenance**.

Impact on Women

- Women spend up to **30 hours per week** on water-related tasks.
- This equates to **4.4 full workdays** (8-hour shifts) per week dedicated to water-related activities.

TIME FOR QUESTIONS

why mexico city?

Geographical

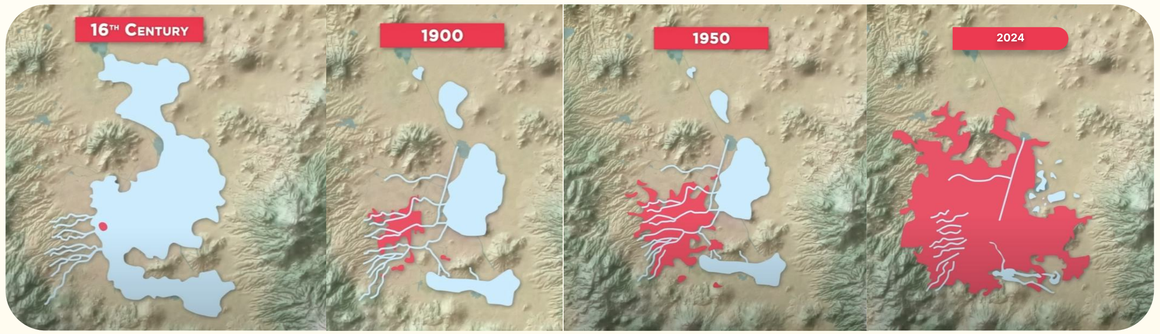
- Mexico City is located in a region where rising global temperatures are expected to intensify droughts. By 2050, climate models predict significant reductions in precipitation during the dry season, while extreme rainfall events may become more frequent, leading to challenges in water storage and flood management.
- The city's reliance on diminishing aquifers makes it particularly vulnerable to long-term changes in rainfall patterns.
- High altitude and mountainous terrain surrounding Mexico City render the importation of water from distant regions an energy-intensive endeavour.
- Over-extraction of groundwater is leading to severe land subsidence (sinking). This not only damages infrastructure but also reduces the capacity of aquifers to recharge effectively in the future.

Social

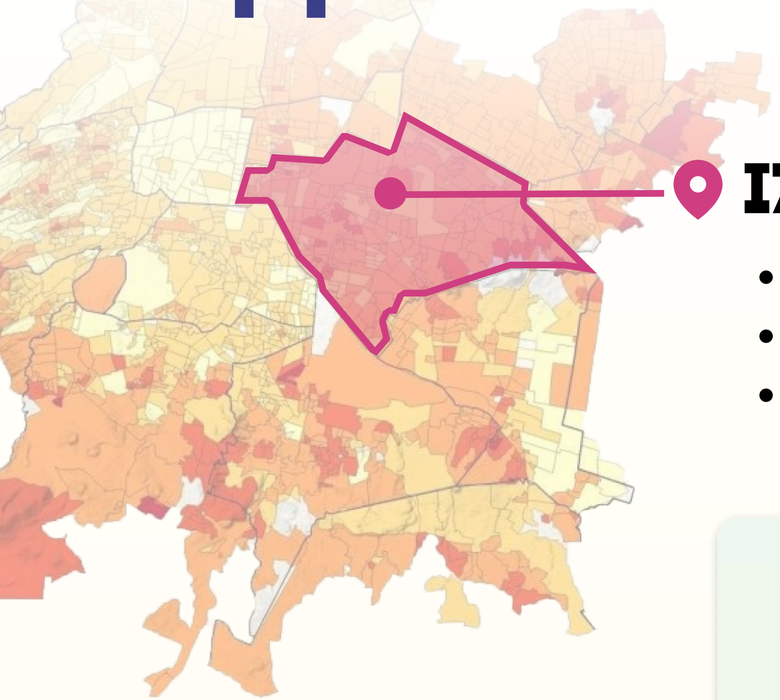
- Mexico City is projected to experience continued urban expansion, adding to the demand for water in both domestic and industrial sectors.
- The increasing density of informal settlements, many of which lack proper water infrastructure, intensifies the crisis and creates significant inequities in water access.
- Wealthier neighbourhoods often have consistent access to water, while marginalized communities rely on irregular and expensive deliveries by water trucks, deepening social divides.
- Access to clean water is already a source of tension, and social unrest could escalate if the scarcity worsens in the future.

Political

- Governance & Fragmentation: Water management in Mexico City involves multiple agencies and overlapping jurisdictions, leading to inefficiencies. Political disputes between local, state, and federal governments further hinder long-term planning and equitable distribution.
- Decades of underinvestment in water infrastructure have left the city with a crumbling network of pipes and reservoirs, with nearly 40% of water lost to leaks.
- Water importation from surrounding basins such as the Lerma and Cutzamala systems has sparked political tension with other states, which view this as exploitation of their resources.
- Efforts to implement large-scale desalination plants, rainwater harvesting, and wastewater recycling face political hurdles, including resistance from vested interests and limited funding.
- Balancing urban growth with water sustainability remains politically contentious, particularly given the need to accommodate a projected population of over 25 million by 2050.



supplementals - users



IZTAPALAPA (SOUTHEAST)

- High level of poverty
- Usage of 40L/day (WHO recommendation 100L/day)
- 40% of household reliance on water truck

PERSONA: SILVA & LUIS

Living with family of 5, the family relies on limited water source and filled water tank is used for all kitchen, laundry and sanitary needs all week.

Central water supply only available on Saturdays

Water bottles and truck supply: economic burden and inefficiency

Water quality and health, hygiene concerns

supplementals - problem

PROBLEM SUMMARY

By 2054, unresolved conflicts and depletion of primary water sources will drive Mexico City residents to seek **cost effective solutions for consistent, reliable water supply**. This challenge can be approached by two angles: water collection and purification. In particular, there is a significant opportunity in **water purification**, which can lessen the dependence on variability of external water sources while integrating into daily life without disrupting user habits.

WHY CAN'T THIS BE SOLVED NOW?

Technical Maturity

Further research on technology needed for effective and stable implementation. It requires significant time for it to be low-cost commercial scale.

Political Coordination

Water management in Mexico City with overlapping responsibilities leads to inefficiency. Expanding current efforts on governmental decentralisation [1], system can settle if the governance is streamlined, which is unlikely to happen in a short term.

Generational and Behavioural Shifts

Along with user's long-term behavioural commitments to conserve and manage water, the system can be implemented more successfully with better sense of ownership.

[1] https://www.oidp.net/en/content.php?id=1917#:~:text=We%20are%20fortunate%20to%20come,central%20streets%20of%20the%20municipality

supplementals - design objectives

Durability

Withstand the environmental conditions, enabling persistent use with minimal maintenence

- Technical maintenance period without user intervention
- Material Selection

Affordability

System viable for initial setup, can be maintained by low to medium income households in CDMX

- Feasible initial cost based on funding examples
- Low or no maintenance costs

Water Quality

Processed water meets the standard for potable water, consistently

- Contaminants filtered in each stage of the system

■ Design Criteria

■ Evaluation Features

Decentralisation

System operates independently, resilient to disruption in central system

- Decentralised source and flow
- Secure connection to each household

Ease of Use

Minimal user intervention in operation and reduce the current effort in water activities

- Process familiar to users
- Change in user journey

supplementals - financial

MFC COST

Based on the research, MFC system treating 100,000L/day to have $32760/year for 10 year life spans. **$380528** total considering inflation and economic changes. [1]

OUR SYSTEM

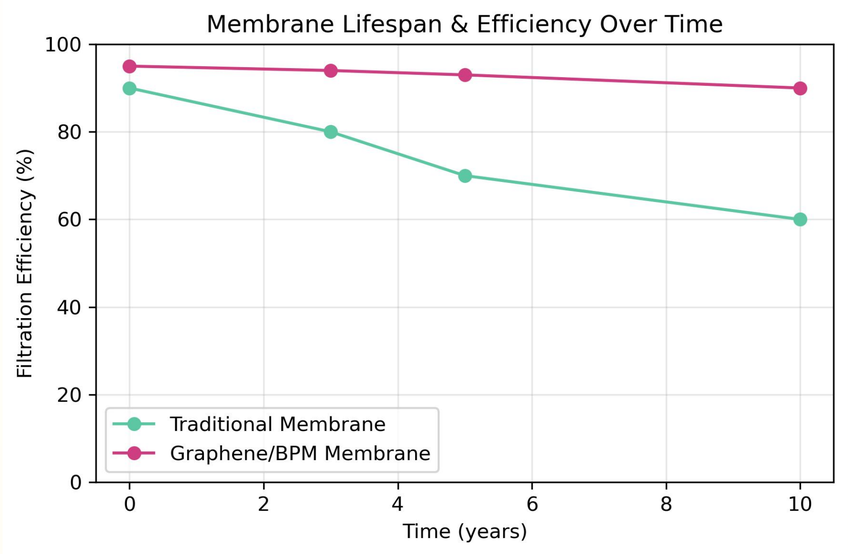
Each unit generates 140L/day, 100000/140 = 714
380528/714 = **$532 per system** (in total)

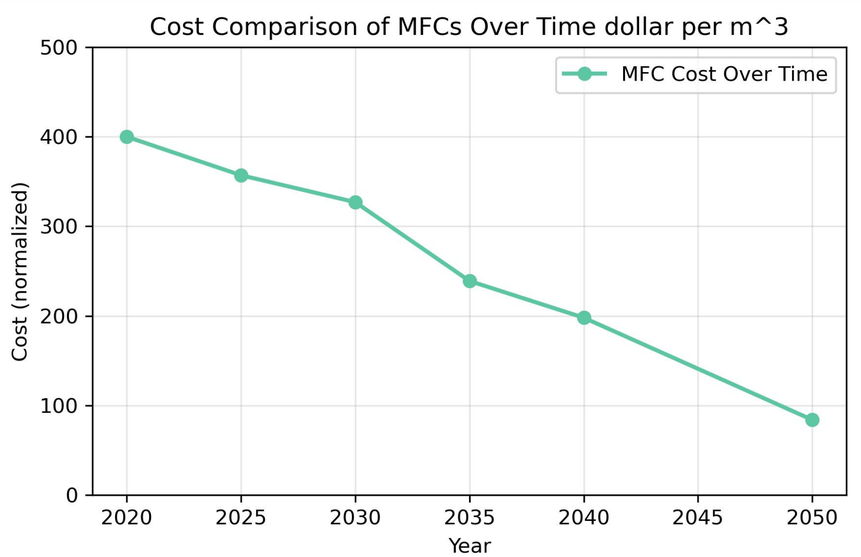
1.8 million residents in Iztapalapa → **$957 million**

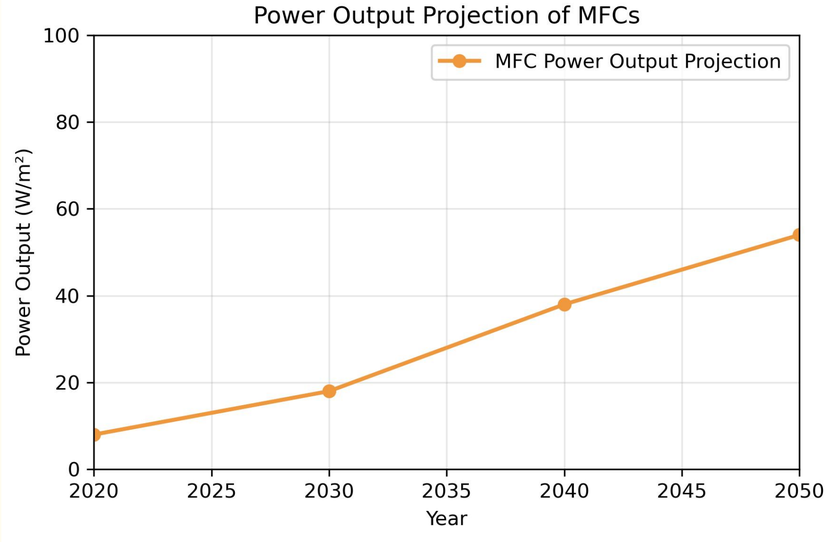
- *(2010) 10 years lifespan → (2054) 50 years unless faulty*
- *Initial cost spread over longer term, resulting in lower cost per year*

[1] chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/pdfdirect/10.1002/elan.200980011

supplementals - technical







supplementals - technical

Optimized Filtration Strategy: Enhancing Efficiency & Sustainability

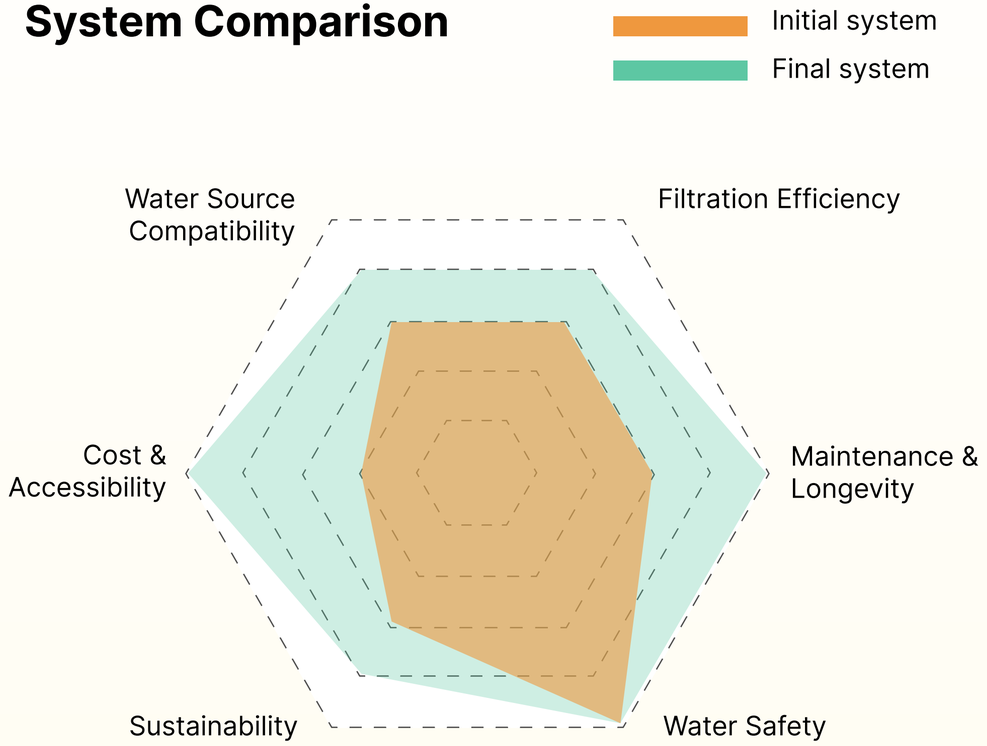
Initial Filtration Strategy (Rainwater Harvesting)

Moss layer (Biological Pre-filtration) → Clay layer (Sediment & Particulate) → BES Tank → Nano-membrane Filtration

Final Filtration Strategy (Grey & Tank water)

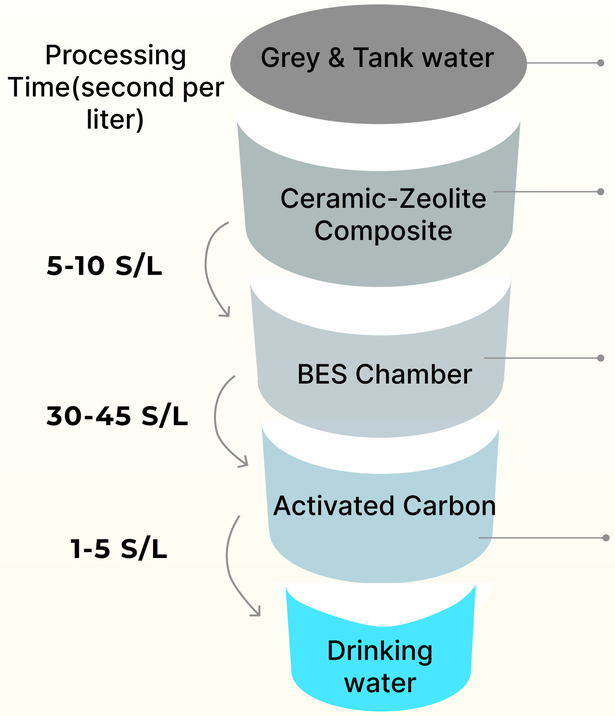
Ceramic-Zeolite Composite Filter → BES Tank with Microbial Fuel Cells → Coconut Shell Activated Carbon

System Comparison



supplementals - technical

Efficiency & Processing Time Breakdown



Initial Water Source – contain suspended solids, organic waste, Arsenic compound, phosphates, and heavy metals.

Removes 85% of sediment & large particulates. Reduces phosphates, and some heavy metals. Lifespan: 5-10 years (Washable & reusable).

Breaks down 95% of organic pollutants and removes 88% of heavy metals, generating 25-50 W/m².

Removes 90% of VOCs, chlorine, & remaining organics. Improves water taste & odor. Lifespan: 2-4 years (Replaceable carbon media).

Reduced Cost & Improved Accessibility

By using reusable **ceramic-zeolite filters and MFC-based BES**, the system operational expenses drop by **40-60%**, while BES self-sustaining energy (25-50 W/m²) lowers external power dependence.

Modular & Easy Maintenance

All filtration components are **detachable and replaceable**, including ceramic-zeolite (5-10 yrs), BES electrodes, and activated carbon (2-4 yrs), ensuring easy maintenance and longevity.

Optimized Water Source Utilization

Compared to rainwater, household greywater provides a **consistent daily input**, increasing filtration system efficiency by 30-50%water output.

pollutants before processing